

Results

01 • GRANGER CAUSALITY

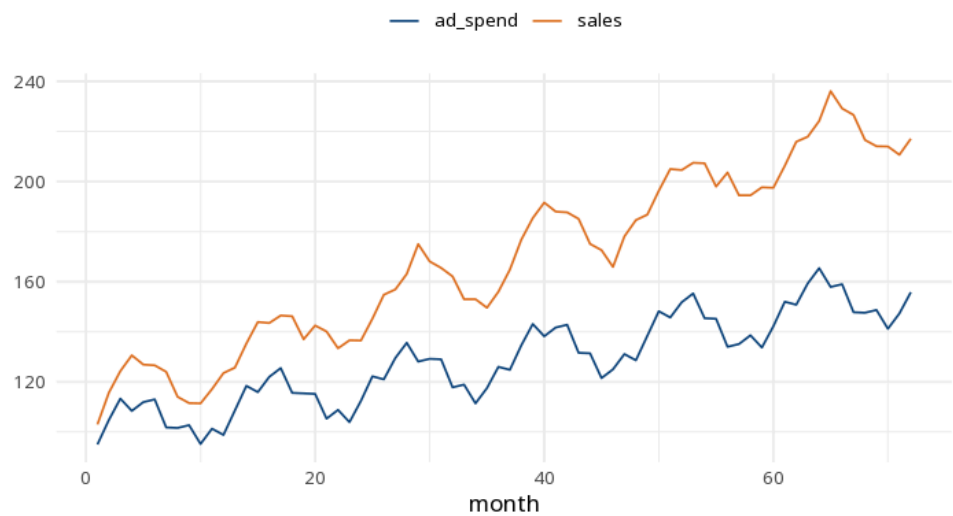
does X predict future Y?

Table. Granger tests (both directions)

Direction	F	df	p
X→Y	47.89	4, 63	<.001
Y→X	30.62	4, 63	<.001

tests predictive precedence, not true causation; both series should be stationary first. Tests use a max lag of 4; select the lag on theory or by AIC/BIC (vars::VARselect) and report it.

Figure. Series together



How to read this test

A significant X→Y p means past values of X help predict Y beyond Y's own past — this is predictive precedence, not proof that X causes Y. Check both directions. The result depends entirely on the lag order (this card uses a max lag of 4): choose it on theory or by an information criterion — vars::VARselect reports the AIC- and BIC-minimising lag — and report it, since different lags can flip the conclusion. Make both series stationary first. Method: R lmtest::grangertest (Granger, 1969). Your significance threshold (α) is 0.05.

APA template: Granger test X→Y: F(4,63)=47.89, p < .001; Y→X: F(4,63)=30.62, p < .001 (lag=4).

